ic_autolab round 2 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T02:11:15Z at git `5840364fa4f38a0c47dcdd7f167c580394813cb1`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**no_challenger**	none (best candidate `mitm2-n22`, train S 11.449 / 12.552 / 22.481)	-	-	-	-	-	8.679 / 9.366 / 15.550	1.340 / 1.430 / 2.245	inc: 6.5x / 6.5x / 6.9x
24	**improvement**	ksum4-n12	0.917 [0.879, 0.957]	0.911 [0.869, 0.955]	0.880 [0.822, 0.942]	sel < inc	6.985 / 7.514 / 12.271	7.613 / 8.246 / 13.944	1.160 / 1.264 / 2.201	6.0x / 5.9x / 5.6x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: no_challenger
- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.9175 [0.879, 0.957] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

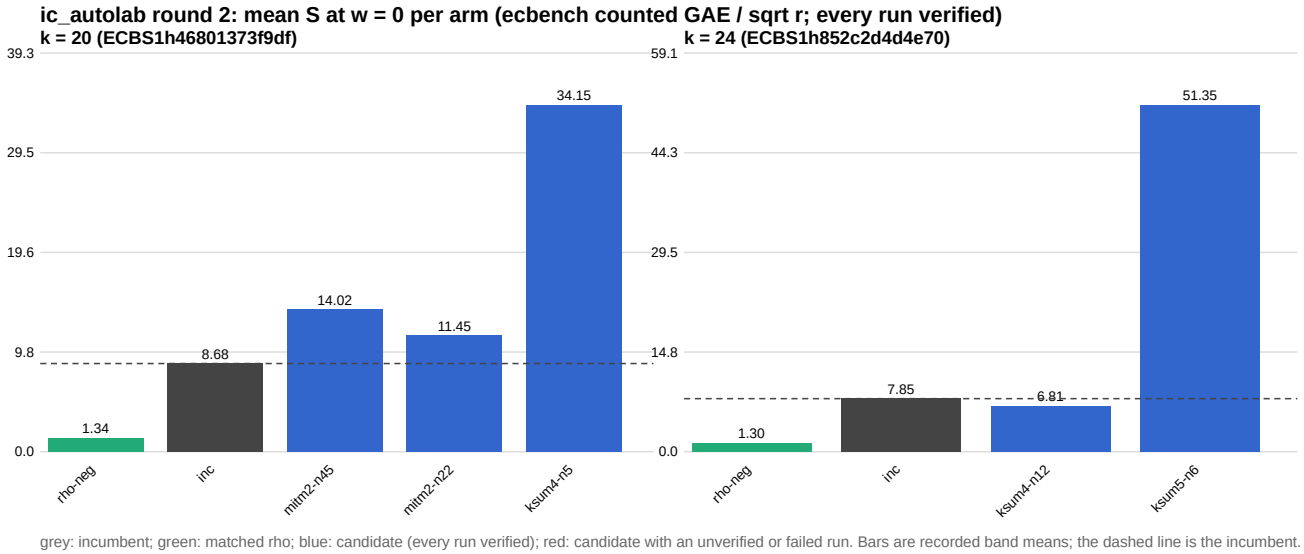
Train (development screen, in sample)

k = 20: session `ECBS1h46801373f9df` (run exit 0, verify exit 0, receipt sha256 `ea315e4fa7d1217043c687740539f26e08f78fca204a19806bed3ed25e234039`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	16	16	1.340 / 1.430 / 2.245	matched reference
`inc`	ksum3	11	`prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=3`	16	16	8.679 / 9.366 / 15.550	incumbent
`mitm2-n45`	mitm2	45	`prime-projected-abscissa:size=45,scan_cap=720`	`mitm:negation_folded=1`	16	16	14.020 / 14.738 / 21.204	alternative mitm2 (best S(0) 11.170 at N = 32): N = 1.4 x 32
`mitm2-n22`	mitm2	22	`prime-projected-abscissa:size=22,scan_cap=352`	`mitm:negation_folded=1`	16	16	11.449 / 12.552 / 22.481	alternative mitm2 (best S(0) 11.170 at N = 32): N = 0.7 x 32
`ksum4-n5`	ksum4	5	`prime-projected-abscissa:size=5,scan_cap=80`	`ksum:m=4`	16	16	34.150 / 39.915 / 91.799	exploration: family ksum4 untried at k = 20, N = 0.15 x N_a
`ksum5-n4`	ksum5	4	`prime-projected-abscissa:size=4,scan_cap=64`	`ksum:m=5`	0	0	no verified runs	exploration: family ksum5 untried at k = 20, N = 0.08 x N_a

k = 24: session `ECBS1h852c2d4d4e70` (run exit 0, verify exit 0, receipt sha256 `32def7afcbada285d3a8d608cbf69567402b52479b8fa9795645119d4b6d955e`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	16	16	1.301 / 1.420 / 2.485	matched reference
inc	ksum3	20 prime-projected-abscissa:size=20,scan_cap=320	ksum:m=3	16	16	7.848 / 8.521 / 14.572	incumbent
ksum4-n12	ksum4	12 prime-projected-abscissa:size=12,scan_cap=192	ksum:m=4	16	16	6.814 / 7.309 / 11.769	exploration: family ksum4 untried at k = 24, N = 0.15 x N_a
ksum5-n6	ksum5	6 prime-projected-abscissa:size=6,scan_cap=96	ksum:m=5	16	16	51.352 / 61.205 / 149.885	exploration: family ksum5 untried at k = 24, N = 0.08 x N_a



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unverified or failed run. Bars are recorded band means; the dashed line is the incumbent.

train screen

Selection

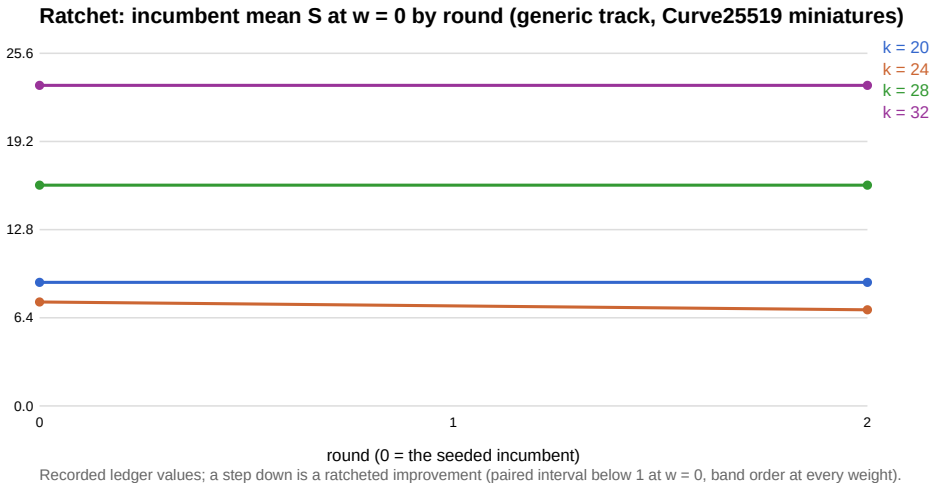
- k = 20: no_challenger (Pareto alternative at w = 1: `mitm2-n45`)
- k = 24: challenger `ksum4-n12` (Pareto alternative at w = 1: `ksum4-n12`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
24	ECBS1h32d89e6cb3e5	0	0	fec1207de337ec6591e1a679a3c12414670f0ca2f5bea33641cc97a1fb699d92	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	ksum3-n11 (prime-projected-abscissa:size=11,scan_cap=176 + ksum:m=3)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	ksum4-n12 (prime-projected-abscissa:size=12,scan_cap=192 + ksum:m=4)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	mitm2-n203 (prime-projected-abscissa:size=203,scan_cap=3248 + mitm:negation_folded=1)	0	16.057 / 17.564 / 31.130	16.6x / 16.6x / 16.7x
32	mitm2-n512 (prime-projected-abscissa:size=512,scan_cap=8192 + mitm:negation_folded=1)	0	23.310 / 25.520 / 45.410	24.7x / 24.6x / 24.3x



ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.